CV5570 – Computer Vision for Construction Engineering

3D Reconstruction



Hands On – Assignment 06

By

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1. Image Set

The image dataset used in this assignment includes 110 pictures of a photo frame taken with a Samsung Galaxy S24 Ultra smartphone. Each picture has a size of 4000 × 3000 pixels (12 megapixels). The pictures were taken from angles to get overlapping visual details for creating a three-dimensional model.

Fig 1 – 6. Sample input images of the physical photo frame captured using a Samsung Galaxy S24 Ultra



The photo frame was selected as the reconstruction object because its physical dimensions could be measured and used to establish the scale of the 3D model. The 110 photographs were processed using COLMAP to estimate camera poses, reconstruct the scene geometry, and generate sparse and dense point clouds.

1.1 Image Acquisition Details

Parameter	Description
Object	Physical photo frame
Camera device	Samsung Galaxy S24 Ultra
Number of images	110
Image resolution	4000 × 3000 pixels
Image size	12 megapixels per image
Physical Height	23 cm
Physical Width	17.5 cm
Reconstruction Software	COLMAP

2. Sparse and Dense Cloud Models

2.1 COLMAP Reconstruction Settings

The following settings were used for automatic reconstruction in COLMAP.

Parameter	Configuration
Data type	Individual images
Quality	Medium
Sparse model	Enabled
Dense model	Enabled
Number of threads	-1
Random seed	-1
Mesher	Poisson
GPU acceleration	Enabled
GPU index	0

The automatic reconstruction workflow was configured to generate both sparse and dense reconstruction outputs. Medium quality was selected for the reconstruction, and GPU acceleration was enabled. Poisson meshing was selected for surface reconstruction.

2.2 Sparse Point-Cloud Reconstruction

Sparse reconstruction is the step in creating a three-dimensional model. It uses matching features from images to figure out where the cameras were positioned and to estimate the 3D locations of points. COLMAP uses a method called Structure-from-Motion or SfM to calculate camera positions and build a version of the scene's geometry.

The sparse reconstruction that comes from the images shows the 3D layout of the scene. It gives us a view of the objects in the photo frame and the space, around them. This early model helps in building a detailed and accurate 3D structure later on.

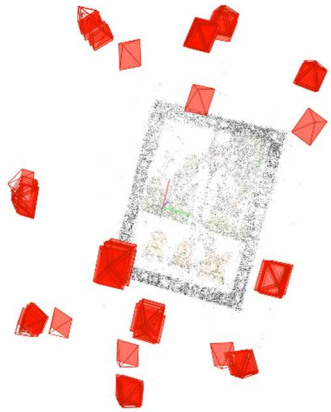


Fig 7. Sparse point-cloud reconstruction of the photo frame generated using COLMAP

2.3 Dense Point-Cloud Reconstruction

Dense reconstruction uses the input images and estimated camera parameters to recover a larger number of three-dimensional points. Compared with the sparse model, the dense point cloud provides a more detailed representation of the visible surfaces of the reconstructed object.

The resulting dense reconstruction in this assignment was then used for scaling, rotating and checking the dimensions.



Fig 8. Dense point-cloud reconstruction generated using COLMAP

3. Camera Calibration

Camera calibration is an important part of three-dimensional reconstruction because it establishes the relationship between the captured images and the reconstructed scene. In this assignment, COLMAP was used to estimate the camera parameters and camera poses from the 110 input photographs.

The calibration results include the intrinsic camera matrix, extrinsic camera parameters, and estimated camera positions. These results were exported to CSV files and a calibration report for further analysis.

3.1 Intrinsic Camera Matrix

The intrinsic camera matrix (K) describes the internal imaging parameters of the camera. It is represented as

$$K = \begin{bmatrix} f_x & s & C_x \\ 0 & f_y & C_y \\ 0 & 0 & 1 \end{bmatrix}$$

where f_x and f_y are the focal lengths in pixel units, C_x and C_y are the principal-point coordinates, and (s) is the skew parameter.

The intrinsic parameters estimated during reconstruction were exported to '*camera_intrinsics.csv.'*

camera id	model	width px	height px	colmap_param_1	colmap_param_2	colmap_param_3	colmap_param_4	fx_px	fy_px	cx_px	cy_px
1	SIMPLE_RADIAL	4000	3000	2819.233	2000	1500	-0.01327	2819.233	2819.233	2000	1500

Table 1. Estimated intrinsic camera parameters from camera_intrinsics.csv

Camera 1:

Model: SIMPLE_RADIAL | Image resolution: 4000 × 3000 pixels

Raw COLMAP parameters: [2819.233007109264, 2000.0, 1500.0, -0.01326505451636744]

2819.23300711	0.00000000	2000.00000000
0.00000000	2819.23300711	1500.00000000
0.00000000	0.00000000	1.00000000

In matrix,

$$K = \begin{bmatrix} 2819.23300711 & 0 & 2000 \\ 0 & 2819.23300711 & 1500 \\ 0 & 0 & 1 \end{bmatrix}$$

Additional distortion parameters: [-0.01326505451636744].

3.2 Extrinsic Camera Parameters

The extrinsic parameters describe the rotation and translation relating the camera coordinate system to the reconstruction coordinate system. They are represented by the rotation matrix (R) and translation vector (t):

$$[R\ |t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

The estimated extrinsic parameters were exported in '*camera_extrinsics.csv*.'

image_id	image_name	camera_id	r11	r12	r13	r21	r22	r23	r31	r32	r33	tx	ty	tz
1	1.jpg	1	-0.98362	0.035955	-0.17664	-0.79234	-0.04427	-0.99808	0.043374	0.240398	-0.17474	0.050484	0.98332	4.668896
2	10.jpg	1	-0.98576	0.057349	-0.15808	-0.68739	-0.06312	-0.9975	0.031716	0.109432	-0.15587	0.041242	0.986917	4.782082
3	100.jpg	1	0.999346	-0.00692	-0.03551	2.12248	0.007367	0.999894	0.012538	-0.30152	0.035415	-0.01279	0.999291	1.296986
4	101.jpg	1	0.999461	-0.00445	-0.03254	2.102798	0.004891	0.999895	0.013632	-0.34929	0.032474	-0.01378	0.999378	1.197683
5	102.jpg	1	0.999352	-0.00571	-0.03553	2.169724	0.005942	0.999963	0.006277	-0.26479	0.03549	-0.00648	0.999349	1.345252
6	103.jpg	1	-0.90741	0.095224	0.409312	-2.98082	-0.05424	-0.99238	0.110618	-0.16255	0.416727	0.078174	0.905664	1.89807
7	104.jpg	1	-0.86803	0.302466	0.393759	-3.03025	-0.33461	-0.94226	-0.01384	0.955908	0.366835	-0.14377	0.919109	1.707876
8	105.jpg	1	-0.99448	-0.01947	-0.10313	-0.58046	0.049199	-0.95447	-0.29423	2.177667	-0.0927	-0.29768	0.950154	2.100808
9	106.jpg	1	-0.98337	0.093655	-0.15558	-0.08097	-0.17421	-0.72833	0.662707	-2.42918	-0.05125	0.678792	0.73254	2.291608
10	107.jpg	1	-0.90629	-0.07029	0.416778	-3.47283	0.196541	-0.94307	0.268318	-1.12019	0.374189	0.325087	0.868505	1.66898
11	108.jpg	1	-0.76789	0.338824	-0.54364	3.372158	-0.43055	-0.90137	0.046373	-0.38702	-0.47431	0.269675	0.838037	1.011622
12	109.jpg	1	-0.80697	0.002559	-0.59058	2.293101	-0.05953	-0.99525	0.077027	0.1859	-0.58758	0.097315	0.803291	1.077895
13	11.jpg	1	-0.99604	-0.02486	-0.08537	-0.69405	0.043646	-0.97318	-0.22587	2.502368	-0.07747	-0.2287	0.970409	1.500384
14	110.jpg	1	-0.82281	-0.31861	-0.47061	3.129463	0.381744	-0.9233	-0.04236	1.473264	-0.42101	-0.2145	0.881326	0.975541
15	12.jpg	1	-0.99625	-0.01131	-0.08578	-0.69274	0.032117	-0.96893	-0.24525	2.467972	-0.08034	-0.24709	0.965657	1.544998
16	13.jpg	1	-0.99643	-0.01576	-0.08295	-0.69348	0.034774	-0.97184	-0.23305	2.464081	-0.07694	-0.2351	0.96892	1.524391
17	14.jpg	1	-0.99687	-0.00594	-0.07887	-0.7033	0.025656	-0.96755	-0.25136	2.418656	-0.07482	-0.25259	0.964676	1.61067
18	15.jpg	1	-0.99709	-0.00091	-0.07623	-0.69288	0.020638	-0.96581	-0.25845	2.380839	-0.07339	-0.25927	0.963013	1.607799
19	16.jpg	1	-0.99673	-0.00741	-0.08052	-0.66787	0.028048	-0.96565	-0.25832	2.43754	-0.07584	-0.25973	0.962698	1.627613
20	17.jpg	1	-0.99722	0.00304	-0.07444	-0.67295	0.017243	-0.96262	-0.27029	2.464622	-0.07248	-0.27082	0.959896	1.635625
21	18.jpg	1	-0.9971	0.001151	-0.07614	-0.69465	0.01869	-0.9656	-0.25935	2.39647	-0.07382	-0.26002	0.962778	1.596239
22	19.jpg	1	-0.99781	0.001658	-0.06612	-0.69291	0.016894	-0.96014	-0.27902	2.383302	-0.06394	-0.27953	0.958006	1.725277
23	2.jpg	1	-0.98331	0.036426	-0.17824	-0.7048	-0.04491	-0.99803	0.043781	0.220831	-0.1763	0.051055	0.983012	4.645955
24	20.jpg	1	-0.9982	-0.00229	-0.05994	-0.66466	0.019349	-0.95812	-0.28572	2.346609	-0.05677	-0.28636	0.956438	1.775211
25	21.jpg	1	-0.98255	0.086222	-0.16482	-0.47024	-0.1636	-0.82223	0.54514	-2.78419	-0.08851	0.56259	0.821984	1.654281
26	22.jpg	1	-0.98259	0.081538	-0.16691	-0.45132	-0.15836	-0.83738	0.523176	-2.80918	-0.09711	0.540501	0.83572	1.650564
27	23.jpg	1	-0.98334	0.085128	-0.16062	-0.45409	-0.15865	-0.83321	0.529714	-2.76258	-0.08874	0.546372	0.832829	1.664291
28	24.jpg	1	-0.98205	0.065162	-0.17702	-0.47481	-0.14732	-0.85106	0.503978	-2.84353	-0.11781	0.521009	0.845381	1.513522
29	25.jpg	1	-0.98219	0.072585	-0.17328	-0.46105	-0.15341	-0.84226	0.516785	-2.80989	-0.10843	0.534167	0.838396	1.599161
30	26.jpg	1	-0.98222	0.056054	-0.17916	-0.50926	-0.1408	-0.85119	0.505627	-2.8882	-0.12415	0.521863	0.843946	1.5196
31	27.jpg	1	-0.98223	0.045839	-0.18198	-0.44507	-0.13376	-0.85119	0.507534	-2.9432	-0.13164	0.522858	0.842194	1.465634
32	28.jpg	1	-0.98229	0.05012	-0.18052	-0.47425	-0.13678	-0.8503	0.50821	-2.94257	-0.12802	0.523902	0.842102	1.491389
33	29.jpg	1	-0.98201	0.046415	-0.18305	-0.40685	-0.13479	-0.85117	0.507287	-2.96442	-0.13226	0.522832	0.842113	1.463996
34	3.jpg	1	-0.98306	0.035123	-0.17989	-0.75386	-0.04523	-0.9976	0.052419	0.236357	-0.17761	0.059668	0.98229	4.686959
35	30.jpg	1	-0.98157	0.057631	-0.18223	-0.35162	-0.14395	-0.85011	0.506542	-2.94957	-0.12572	0.523436	0.842739	1.521061
36	31.jpg	1	-0.83413	-0.08016	-0.54572	3.376116	-0.03246	-0.98053	0.193644	-0.1497	-0.55062	0.179236	0.815289	1.744231
37	32.jpg	1	-0.83514	-0.08443	-0.54352	3.36356	-0.02876	-0.98009	0.196441	-0.15158	-0.54929	0.179688	0.816086	1.747707
38	33.jpg	1	-0.83461	-0.07482	-0.54574	3.383923	-0.0323	-0.98238	0.184073	-0.20258	-0.54989	0.171255	0.817489	1.796921
39	34.jpg	1	-0.83404	-0.07516	-0.54656	3.353743	-0.03389	-0.98183	0.186734	-0.22135	-0.55066	0.174269	0.816338	1.7815
40	35.jpg	1	-0.83295	-0.08036	-0.54748	3.40362	-0.0262	-0.98256	0.184091	-0.23961	-0.55272	0.167685	0.816319	1.811174
41	36.jpg	1	-0.83274	-0.08119	-0.54768	3.38061	-0.02107	-0.98383	0.17788	-0.22478	-0.55327	0.159665	0.817559	1.817125
42	37.jpg	1	-0.83066	-0.08179	-0.55074	3.38152	-0.02247	-0.98342	0.179935	-0.21039	-0.55633	0.161842	0.815049	1.818403
43	38.jpg	1	-0.83273	-0.08338	-0.54736	3.343385	-0.01909	-0.98368	0.178892	-0.2454	-0.55335	0.159421	0.817551	1.827917
44	39.jpg	1	-0.83329	-0.07995	-0.54703	3.38606	-0.01939	-0.98465	0.173446	-0.20625	-0.5525	0.155136	0.818946	1.799446
45	4.jpg	1	-0.98403	0.036037	-0.17434	-0.74406	-0.04586	-0.99756	0.052667	0.207781	-0.17201	0.059822	0.983277	4.712539
46	40.jpg	1	-0.83436	-0.08728	-0.54426	3.364298	-0.01249	-0.98414	0.176957	-0.24357	-0.55107	0.154443	0.820041	1.820243
47	41.jpg	1	-0.92252	0.072712	0.379032	-3.46198	-0.02934	-0.99246	0.118989	0.288775	0.384827	0.098651	0.917702	2.469945
48	42.jpg	1	-0.92575	0.072532	0.371126	-3.4991	-0.03024	-0.99249	0.118534	0.286093	0.376936	0.098509	0.920986	2.360075
49	43.jpg	1	-0.91898	0.077837	0.386557	-3.43115	-0.03743	-0.99312	0.110997	0.208085	0.392535	0.087536	0.915562	2.449947
50	44.jpg	1	-0.92105	0.074919	0.382164	-3.46103	-0.03318	-0.99285	0.114675	0.261274	0.388023	0.092942	0.916952	2.496843
51	45.jpg	1	-0.91841	0.090562	0.385123	-3.41664	-0.05146	-0.99252	0.110684	0.181886	0.392267	0.081836	0.916204	2.404684
52	46.jpg	1	-0.91656	0.082185	0.391372	-3.40071	-0.04603	-0.99383	0.100908	0.179434	0.397251	0.074475	0.914683	2.473598
53	47.jpg	1	-0.91219	0.088362	0.400134	-3.35724	-0.05766	-0.99444	0.088165	0.04123	0.405698	0.057353	0.912206	2.502519
54	48.jpg	1	-0.9133	0.079835	0.399388	-3.36518	-0.04471	-0.99433	0.096509	0.141903	0.404827	0.070284	0.911688	2.562814
55	49.jpg	1	-0.90772	0.093663	0.408987	-3.31496	-0.06309	-0.99415	0.087648	0.102551	0.414805	0.053757	0.908321	2.630942
56	5.jpg	1	-0.98372	0.037657	-0.17572	-0.75373	-0.04768	-0.99745	0.053188	0.205075	-0.17327	0.060701	0.983002	4.729403
57	50.jpg	1	-0.90563	0.090899	0.414203	-3.31215	-0.06354	-0.99482	0.079388	0.100512	0.419273	0.045577	0.906716	2.635693
58	51.jpg	1	-0.83567	0.394869	0.38176	-3.58031	-0.42294	-0.90609	0.011397	0.845194	0.350408	-0.15194	0.924191	1.488015
59	52.jpg	1	-0.83307	0.38789	0.394383	-3.53541	-0.41877	-0.90805	0.008518	0.947483	0.361424	-0.15806	0.918907	1.591453
60	53.jpg	1	-0.83965	0.376434	0.391519	-3.53591	-0.41314	-0.91061	-0.01048	0.972903	0.352575	-0.17055	0.92011	1.576051
61	54.jpg	1	-0.83736	0.380342	0.392639	-3.52468	-0.4163	-0.9092	-0.0071	0.951096	0.354285	-0.1694	0.919665	1.569933
62	55.jpg	1	-0.84026	0.378003	0.388687	-3.5652	-0.41781	-0.90832	-0.01987	1.012663	0.345541	-0.17909	0.921156	1.581398

63	56.jpg	1	-0.84041	0.376747	0.389571	-3.56829	-0.4153	-0.90954	-0.01633	0.998113	0.348177	-0.17551	0.920852	1.589662
64	57.jpg	1	-0.83328	0.392415	0.389435	-3.50778	-0.43664	-0.8992	-0.0282	0.856672	0.339111	-0.19354	0.920622	1.618807
65	58.jpg	1	-0.83995	0.378704	0.38868	-3.56758	-0.42029	-0.90706	-0.02448	1.007086	0.343284	-0.18392	0.921048	1.538924
66	59.jpg	1	-0.84739	-0.21754	0.484361	-3.63396	0.379932	-0.88567	0.266924	-0.83668	0.370917	0.410213	0.833154	2.144452
67	6.jpg	1	-0.98411	0.045002	-0.17176	-0.77313	-0.05354	-0.99753	0.045428	0.171744	-0.16929	0.053903	0.984091	4.742564
68	60.jpg	1	-0.84596	-0.2134	0.488677	-3.58492	0.368688	-0.89616	0.246901	-0.91115	0.385245	0.389039	0.8368	2.196446
69	61.jpg	1	-0.84105	-0.21092	0.498143	-3.53168	0.357194	-0.9081	0.21857	-0.95137	0.40626	0.361762	0.839095	2.254222
70	62.jpg	1	-0.84693	-0.20596	0.490191	-3.55415	0.355848	-0.90458	0.23475	-0.95453	0.395068	0.373251	0.839407	2.193521
71	63.jpg	1	-0.83714	-0.2015	0.508515	-3.51447	0.349874	-0.91188	0.214638	-0.96734	0.420452	0.357599	0.833872	2.338585
72	64.jpg	1	-0.83528	-0.19592	0.51373	-3.46325	0.344703	-0.91453	0.211675	-1.01684	0.428352	0.353892	0.83143	2.381994
73	65.jpg	1	-0.82934	-0.2057	0.51951	-3.42261	0.352718	-0.91383	0.201248	-0.92007	0.43335	0.350143	0.830427	2.47293
74	66.jpg	1	-0.82968	-0.19524	0.522991	-3.4003	0.342444	-0.91787	0.200609	-1.02104	0.440873	0.345535	0.828394	2.566194
75	67.jpg	1	0.265587	-0.47421	0.839396	-3.6821	0.681421	0.70825	0.18452	-0.5333	-0.682	0.522976	0.511241	2.605998
76	68.jpg	1	0.270241	-0.45112	0.850566	-3.64642	0.689652	0.707152	0.15594	-0.48384	-0.67183	0.544453	0.502216	2.649261
77	69.jpg	1	0.274095	-0.45568	0.846893	-3.65984	0.686518	0.709401	0.15951	-0.47341	-0.67347	0.537687	0.507276	2.578969
78	7.jpg	1	-0.985	0.051944	-0.16454	-0.80297	-0.06006	-0.99719	0.044756	0.140573	-0.16175	0.053967	0.985355	4.781923
79	70.jpg	1	0.267949	-0.42208	0.866053	-3.5399	0.692604	0.709252	0.13138	-0.40744	-0.6697	0.564629	0.482381	3.034647
80	71.jpg	1	0.26953	-0.43672	0.858273	-3.57008	0.692379	0.707327	0.142477	-0.4571	-0.6693	0.555848	0.493019	2.825602
81	72.jpg	1	0.263934	-0.41861	0.868969	-3.5494	0.693112	0.708836	0.130947	-0.43951	-0.67077	0.567731	0.477228	3.164298
82	73.jpg	1	0.261324	-0.41682	0.870617	-3.54133	0.693863	0.708136	0.130757	-0.4234	-0.67102	0.569919	0.474266	3.223101
83	74.jpg	1	0.265655	-0.41634	0.869532	-3.55846	0.692452	0.70995	0.128379	-0.40309	-0.67077	0.568004	0.4769	3.212486
84	75.jpg	1	0.63585	0.078545	0.767806	-3.83783	-0.121	0.992652	-0.00134	-1.77308	-0.76227	-0.09205	0.640681	1.645049
85	76.jpg	1	0.623493	0.083679	0.777338	-3.77404	-0.10944	0.993808	-0.0192	-1.82056	-0.77413	-0.07309	0.62879	1.708662
86	77.jpg	1	0.620407	0.090811	0.779005	-3.77833	-0.12542	0.991979	-0.01575	-1.72671	-0.77419	-0.08793	0.62682	1.772824
87	78.jpg	1	0.620467	0.094436	0.778526	-3.78204	-0.10921	0.993455	-0.03347	-1.79621	-0.77659	-0.06426	0.626719	1.669332
88	79.jpg	1	0.62001	0.095335	0.77878	-3.75377	-0.11341	0.993056	-0.03128	-1.76238	-0.77635	-0.06893	0.626517	1.700509
89	8.jpg	1	-0.98626	0.05476	-0.15586	-0.76732	-0.06001	-0.99777	0.029195	0.119907	-0.15392	0.038148	0.987347	4.823448
90	80.jpg	1	0.617311	0.09892	0.780475	-3.78804	-0.10425	0.9936	-0.04347	-1.77313	-0.77978	-0.05453	0.623673	1.659012
91	81.jpg	1	0.615806	0.100685	0.781438	-3.75347	-0.11117	0.992982	-0.04033	-1.70221	-0.78002	-0.06204	0.622678	1.691627
92	82.jpg	1	0.617379	0.098558	0.780467	-3.75838	-0.10358	0.993667	-0.04354	-1.75962	-0.77982	-0.05396	0.623678	1.654251
93	83.jpg	1	0.604505	0.01092	0.796527	-3.72535	0.083594	0.993516	-0.07706	-0.26737	-0.7922	0.11317	0.599672	2.154498
94	84.jpg	1	0.591345	0.018411	0.806208	-3.71676	0.075824	0.994041	-0.07832	-0.17497	-0.80285	0.107442	0.586426	2.39307
95	85.jpg	1	0.60239	0.015943	0.798042	-3.76305	0.083439	0.993065	-0.08282	-0.26292	-0.79383	0.116479	0.596883	2.26239
96	86.jpg	1	0.591062	0.017172	0.806443	-3.7557	0.078296	0.993831	-0.07855	-0.2491	-0.80282	0.109568	0.586071	2.325037
97	87.jpg	1	0.588998	0.018421	0.807924	-3.74869	0.075498	0.994113	-0.07771	-0.20556	-0.8046	0.106766	0.58414	2.405461
98	88.jpg	1	0.590354	0.018835	0.806925	-3.77714	0.077046	0.993848	-0.07957	-0.21085	-0.80346	0.109142	0.58527	2.350849
99	89.jpg	1	0.589708	0.014354	0.807489	-3.77128	0.076961	0.994293	-0.07388	-0.23733	-0.80394	0.105712	0.585238	2.323787
100	9.jpg	1	-0.98658	0.057997	-0.15264	-0.72836	-0.06343	-0.99751	0.030972	0.058016	-0.15047	0.040238	0.987796	4.841258
101	90.jpg	1	0.588095	0.012999	0.808687	-3.75337	0.079358	0.994119	-0.07369	-0.2377	-0.80489	0.107513	0.583605	2.336382
102	91.jpg	1	0.587106	0.009678	0.809452	-3.792	0.082928	0.993949	-0.07203	-0.28505	-0.80525	0.109417	0.582751	2.373511
103	92.jpg	1	0.583615	0.001494	0.812029	-3.76308	0.088862	0.993875	-0.06569	-0.33191	-0.80715	0.110499	0.579908	2.408009
104	93.jpg	1	0.999205	-0.00966	-0.03867	2.28921	0.009639	0.999953	-0.00087	-0.42244	0.038673	0.000495	0.999252	1.296963
105	94.jpg	1	0.999535	0.00022	-0.03049	2.262351	-0.00036	0.999989	-0.00464	-0.4069	0.030491	0.004645	0.999524	1.359129
106	95.jpg	1	0.999468	-0.00345	-0.03242	2.249088	0.003478	0.999994	0.000748	-0.35184	0.032414	-0.00086	0.999474	1.313483
107	96.jpg	1	0.999301	-0.00335	-0.03722	2.314011	0.0033	0.999993	-0.00146	-0.36072	0.037228	0.001332	0.999306	1.319148
108	97.jpg	1	0.99925	-0.00083	-0.03872	2.149933	0.000956	0.999994	0.003228	-0.36855	0.038717	-0.00326	0.999245	1.276292
109	98.jpg	1	0.999216	-0.00478	-0.0393	2.1426	0.004901	0.999983	0.00309	-0.3745	0.039282	-0.00328	0.999223	1.254779
110	99.jpg	1	0.999171	-0.00432	-0.04049	2.123402	0.00448	0.999982	0.003945	-0.3464	0.040468	-0.00412	0.999172	1.197699

Table 2. Estimated camera extrinsic parameters from camera_extrinsics.csv

Image 1: 1.jpg

Camera ID: 1

-0.98361929	0.03595518	-0.17663611	-0.79234233
-0.04427267	-0.99807747	0.04337393	0.24039783
-0.17473700	0.05048359	0.98332008	4.66889581

Camera centre $C = -R^T t$ (COLMAP world coordinates): [0.04710872 0.03272184 -4.74140228]

Image 2: 10.jpg

Camera ID: 1

-0.98575920	0.05734927	-0.15808184	-0.68739112
-0.06311862	-0.99750194	0.03171621	0.10943170
-0.15586804	0.04124246	0.98691652	4.78208249

Camera centre $C = -R^T t$ (COLMAP world coordinates): [0.0746789 -0.04864511 -4.83165101]

Image 3: 100.jpg

Camera ID: 1

0.99934555	-0.00691798	-0.03550501	2.12247991
0.00736725	0.99989425	0.01253832	-0.30152228
0.03541451	-0.01279169	0.99929084	1.29698616

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.16480161 0.33276432 -1.21692715]

Image 4: 101.jpg

Camera ID: 1

0.99946063	-0.00444502	-0.03253764	2.10279776
0.00489074	0.99989512	0.01363194	-0.34928682
0.03247363	-0.01378372	0.99937754	1.19768310

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.13884841 0.3751057 -1.12375607]

Image 5: 102.jpg

Camera ID: 1

0.99935237	-0.00571489	-0.03552721	2.16972408
0.00594154	0.99996265	0.00627722	-0.26479162
0.03549001	-0.00648424	0.99934900	1.34525230

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.21448865 0.28590442 -1.26563014]

Image 6: 103.jpg

Camera ID: 1

-0.90741186	0.09522378	0.40931180	-2.98082010
-0.05424324	-0.99238161	0.11061827	-0.16254967
0.41672699	0.07817393	0.90566409	1.89807018

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.50462581 -0.02584598 -0.48094821]

Image 7: 104.jpg

Camera ID: 1

-0.86802544	0.30246568	0.39375925	-3.03024705
-0.33460957	-0.94225521	-0.01384034	0.95590791
0.36683547	-0.14376938	0.91910941	1.70787550

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.93698492 2.06279514 -0.36330655]

Image 8: 105.jpg

Camera ID: 1

-0.99447755	-0.01947076	-0.10312754	-0.58045612
0.04919933	-0.95446680	-0.29423215	2.17766673
-0.09270289	-0.29768107	0.95015375	2.10080850

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.48963931 2.6925796 -1.41521252]

Image 9: 106.jpg

Camera ID: 1

-0.98337376	0.09365457	-0.15557917	-0.08096659
-0.17421162	-0.72833391	0.66270659	-2.42917581
-0.05124808	0.67879196	0.73254017	2.29160798

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.38537055 -3.31719332 -0.08146081]

Image 10: 107.jpg

Camera ID: 1

-0.90628620	-0.07029480	0.41677807	-3.47283040
0.19654055	-0.94306789	0.26831840	-1.12018764
0.37418863	0.32508706	0.86850520	1.66897993

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.55172929 -1.84309867 0.29844877]

Image 11: 108.jpg

Camera ID: 1

-0.76789010	0.33882399	-0.54363876	3.37215785
-0.43055286	-0.90137325	0.04637343	-0.38701672
-0.47430900	0.26967492	0.83803723	1.01162216

Camera centre C = $-R^T t$ (COLMAP world coordinates): [2.90263697 -1.76422362 1.00340596]

Image 12: 109.jpg

Camera ID: 1

-0.80697195	0.00255904	-0.59058422	2.29310103
-0.05952816	-0.99525037	0.07702655	0.18589978
-0.58758206	0.09731466	0.80329147	1.07789521

Camera centre C = $-R^T t$ (COLMAP world coordinates): [2.49488636 0.07425368 0.47408604]

Image 13: 11.jpg

Camera ID: 1

-0.99603894	-0.02485620	-0.08537333	-0.69405118
0.04364577	-0.97317898	-0.22587101	2.50236759

-0.07746923	-0.22870251	0.97040903	1.50038358

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.6842862 2.76114156 -0.95002693]

Image 14: 110.jpg

Camera ID: 1

-0.82281184	-0.31860817	-0.47060548	3.12946319
0.38174438	-0.92329683	-0.04235794	1.47326392
-0.42101296	-0.21450361	0.88132644	0.97554120

Camera centre C = $-R^T t$ (COLMAP world coordinates): [2.42326463 2.56658956 0.67537669]

Image 15: 12.jpg

Camera ID: 1

-0.99624971	-0.01130991	-0.08578228	-0.69274287
0.03211727	-0.96892703	-0.24525272	2.46797153
-0.08034299	-0.24708804	0.96565651	1.54499798

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.64527963 2.76519999 -0.94608571]

Image 16: 13.jpg

Camera ID: 1

-0.99642925	-0.01576220	-0.08294758	-0.69348079
0.03477363	-0.97184241	-0.23305176	2.46408126
-0.07693857	-0.23510398	0.96892031	1.52439140

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.65940509 2.74215837 -0.96027788]

Image 17: 14.jpg

Camera ID: 1

-0.99686720	-0.00594360	-0.07886990	-0.70329983
0.02565560	-0.96755446	-0.25135664	2.41865646
-0.07481696	-0.25259264	0.96467579	1.61066973

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.64264321 2.74284504 -1.00129793]

Image 18: 15.jpg

Camera ID: 1

-0.99708997	-0.00090809	-0.07622845	-0.69287810
0.02063802	-0.96580527	-0.25844583	2.38083927
-0.07338714	-0.25926695	0.96301349	1.60779851

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.62200587 2.71564694 -0.98583068]

Image 19: 16.jpg

Camera ID: 1

-0.99672508	-0.00740925	-0.08052468	-0.66786888
0.02804750	-0.96565300	-0.25831689	2.43753991
-0.07584496	-0.25972944	0.96269837	1.62761284

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.61060234 2.7716083 -0.99102243]

Image 20: 17.jpg

Camera ID: 1

-0.99722061	0.00303984	-0.07444333	-0.67295472
0.01724311	-0.96262407	-0.27029165	2.46462163
-0.07248258	-0.27082404	0.95989615	1.63562491

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.59502774 2.81751633 -0.95396039]

Image 21: 18.jpg

Camera ID: 1

-0.99709649	0.00115060	-0.07613982	-0.69464969
0.01869003	-0.96560280	-0.25934903	2.39647045
-0.07381924	-0.26001906	0.96277765	1.59623934

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.6195897 2.7298905 -0.96819178]

Image 22: 19.jpg

Camera ID: 1

-0.99781043	0.00165792	-0.06611810	-0.69291292
0.01689358	-0.96013607	-0.27902212	2.38330246
-0.06394497	-0.27952815	0.95800577	1.72527721

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.62133544 2.77170699 -1.0336455]

Image 23: 2.jpg

Camera ID: 1

-0.98331180	0.03642642	-0.17824427	-0.70480000
-0.04490787	-0.99803131	0.04378110	0.22083091
-0.17629858	0.05105504	0.98301180	4.64595535

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [0.13595421 0.00887008 -4.7023237]

Image 24: 20.jpg

Camera ID: 1

-0.99819966	-0.00228539	-0.05993511	-0.66466404
0.01934905	-0.95811839	-0.28571797	2.34660867

-0.05677196	-0.28636327	0.95643767	1.77521113

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.60808985 2.75516517 -1.06724725]

Image 25: 21.jpg

Camera ID: 1

-0.98254840	0.08622249	-0.16481603	-0.47023583
-0.16359738	-0.82222749	0.54514021	-2.78418616
-0.08851292	0.56259011	0.82198408	1.65428112

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.77108978 -3.1793717 0.08047668]

Image 26: 22.jpg

Camera ID: 1

-0.98259490	0.08153813	-0.16690953	-0.45132204
-0.15835786	-0.83738285	0.52317564	-2.80918027
-0.09710842	0.54050116	0.83572032	1.65056395

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.72803887 -3.20769115 0.01495491]

Image 27: 23.jpg

Camera ID: 1

-0.98333856	0.08512805	-0.16061911	-0.45408835
-0.15865484	-0.83320540	0.52971445	-2.76258347
-0.08873515	0.54637164	0.83282897	1.66429146

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.73713866 -3.17246547 0.00437497]

Image 28: 24.jpg

Camera ID: 1

-0.98204773	0.06516215	-0.17702021	-0.47481260
-0.14731595	-0.85106033	0.50397849	-2.84353360
-0.11781455	0.52100883	0.84538129	1.51352230

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.70687155 -3.17763733 0.0695249]

Image 29: 25.jpg

Camera ID: 1

-0.98219459	0.07258535	-0.17327767	-0.46104542
-0.15341442	-0.84225737	0.51678480	-2.80988720
-0.10843339	0.53416653	0.83839628	1.59916143

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.71051103 -3.18740158 0.03148714]

Image 30: 26.jpg

Camera ID: 1

-0.98222256	0.05605405	-0.17915575	-0.50925766
-0.14080143	-0.85118534	0.50562682	-2.88820141
-0.12415232	0.52186346	0.84394593	1.51959991

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.71820539 -3.2228724 0.08665551]

Image 31: 27.jpg

Camera ID: 1

-0.98223277	0.04583926	-0.18198229	-0.44507162
-0.13375651	-0.85118621	0.50753446	-2.94319931
-0.13163581	0.52285830	0.84219429	1.46563379

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.63790611 -3.2511277 0.17843153]

Image 32: 28.jpg

Camera ID: 1

-0.98229409	0.05011984	-0.18051683	-0.47424947
-0.13677924	-0.85030246	0.50820976	-2.94257019
-0.12802251	0.52390240	0.84210244	1.49138910

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.67740359 -3.2596477 0.15393049]

Image 33: 29.jpg

Camera ID: 1

-0.98200800	0.04641506	-0.18304628	-0.40684998
-0.13478925	-0.85117090	0.50728686	-2.96442354
-0.13225791	0.52283243	0.84211288	1.46399580

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-0.60547732 -3.26977155 0.19649103]

Image 34: 3.jpg

Camera ID: 1

-0.98306028	0.03512346	-0.17988560	-0.75385641
-0.04523485	-0.99760015	0.05241894	0.23635663
-0.17761277	0.05966808	0.98228989	4.68695883

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [0.10206899 -0.01739437 -4.75194974]

Image 35: 30.jpg

Camera ID: 1

-0.98156643	0.05763055	-0.18222531	-0.35162245

-0.14395082	-0.85011377	0.50654194	-2.94956636
-0.12571996	0.52343605	0.84273910	1.52106085

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.57850558 -3.28338086 0.14814711]

Image 36: 31.jpg

Camera ID: 1

-0.83412779	-0.08016142	-0.54571510	3.37611578
-0.03245727	-0.98053477	0.19364421	-0.14969975
-0.55061543	0.17923644	0.81528948	1.74423118

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.77165376 -0.18878135 0.44933252]

Image 37: 32.jpg

Camera ID: 1

-0.83513848	-0.08442957	-0.54352127	3.36356030
-0.02876273	-0.98009373	0.19644081	-0.15157700
-0.54928720	0.17968844	0.81608556	1.74770729

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.7646721 -0.17861851 0.43166379]

Image 38: 33.jpg

Camera ID: 1

-0.83460961	-0.07481922	-0.54573700	3.38392298
-0.03229607	-0.98238178	0.18407337	-0.20258257
-0.54989432	0.17125457	0.81748891	1.79692052

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.8058284 -0.25356178 0.41505945]

Image 39: 34.jpg

Camera ID: 1

-0.83404358	-0.07515678	-0.54655537	3.35374321
-0.03389455	-0.98182566	0.18673384	-0.22135310
-0.55065640	0.17426941	0.81633798	1.78150045

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.77065994 -0.27573465 0.42003399]

Image 40: 35.jpg

Camera ID: 1

-0.83295190	-0.08036127	-0.54747895	3.40361976
-0.02620348	-0.98255984	0.18409109	-0.23960634
-0.55272462	0.16768488	0.81631935	1.81117428

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.82985364 -0.2656149 0.42902295]

Image 41: 36.jpg

Camera ID: 1

-0.83273714	-0.08119271	-0.54768294	3.38060953
-0.02106609	-0.98382660	0.17788041	-0.22478485
-0.55326764	0.15966516	0.81755853	1.81712467

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.81578004 -0.23679997 0.4058812]

Image 42: 37.jpg

Camera ID: 1

-0.83065781	-0.08178567	-0.55074377	3.38151979
-0.02247407	-0.98342174	0.17993498	-0.21038952
-0.55632950	0.16184185	0.81504889	1.81840337

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.81578896 -0.22463555 0.41811975]

Image 43: 38.jpg

Camera ID: 1

-0.83273026	-0.08337862	-0.54736489	3.34338469
-0.01909494	-0.98368343	0.17889182	-0.24540479
-0.55334953	0.15942053	0.81755085	1.82791675

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.79092848 -0.25404129 0.37953741]

Image 44: 39.jpg

Camera ID: 1

-0.83328524	-0.07995125	-0.54703154	3.38605959
-0.01938829	-0.98465260	0.17344553	-0.20624772
-0.55250322	0.15513561	0.81894636	1.79944625

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.81175453 -0.21152086 0.4144042]

Image 45: 4.jpg

Camera ID: 1

-0.98402651	0.03603741	-0.17433627	-0.74406290
-0.04586382	-0.99755836	0.05266717	0.20778078
-0.17201261	0.05982162	0.98327668	4.71253892

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [0.08796813 -0.04782415 -4.77439001]

Image 46: 40.jpg

Camera ID: 1

-0.83436489	-0.08727608	-0.54425924	3.36429794
-0.01248710	-0.98413932	0.17695725	-0.24356855
-0.55107105	0.15444314	0.82004147	1.82024306

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.80709388 -0.22720671 0.38147665]

Image 47: 41.jpg

Camera ID: 1

-0.92252243	0.07271199	0.37903209	-3.46197763
-0.02933612	-0.99246209	0.11898904	0.28877544
0.38482691	0.09865073	0.91770163	2.46994499

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-4.13578177 0.29466411 -0.98883304]

Image 48: 42.jpg

Camera ID: 1

-0.92574546	0.07253216	0.37112590	-3.49910419
-0.03024182	-0.99248932	0.11853434	0.28609278
0.37693604	0.09850911	0.92098598	2.36007489

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-4.12022514 0.30525274 -0.90889952]

Image 49: 43.jpg

Camera ID: 1

-0.91897514	0.07783721	0.38655668	-3.43115173
-0.03742710	-0.99311570	0.11099738	0.20808534
0.39253524	0.08753613	0.91556186	2.44994724

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-4.10704572 0.25926519 -0.93984054]

Image 50: 44.jpg

Camera ID: 1

-0.92105244	0.07491890	0.38216432	-3.46103362
-0.03317777	-0.99284888	0.11467496	0.26127418
0.38802274	0.09294229	0.91695152	2.49684337

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-4.14795696 0.28664026 -0.99676236]

Image 51: 45.jpg

Camera ID: 1

-0.91841107	0.09056174	0.38512293	-3.41664498
-0.05145607	-0.99252276	0.11068352	0.18188619
0.39226697	0.08183606	0.91620384	2.40468378

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-4.07180345 0.29315365 -0.90748399]

Image 52: 46.jpg

Camera ID: 1

-0.91655504	0.08218548	0.39137246	-3.40070576
-0.04602635	-0.99383051	0.10090838	0.17943440
0.39725109	0.07447464	0.91468306	2.47359776

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-4.09131471 0.27359573 -0.94972182]

Image 53: 47.jpg

Camera ID: 1

-0.91218678	0.08836164	0.40013435	-3.35723534
-0.05765522	-0.99443597	0.08816453	0.04122991
0.40569835	0.05735269	0.91220585	2.50251944

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-4.07531659 0.1941251 -0.94310271]

Image 54: 48.jpg

Camera ID: 1

-0.91329947	0.07983465	0.39938766	-3.36518489
-0.04471389	-0.99432723	0.09650921	0.14190302
0.40482680	0.07028363	0.91168825	2.56281359

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-4.10457219 0.22963253 -1.00616868]

Image 55: 49.jpg

Camera ID: 1

-0.90772063	0.09366273	0.40898723	-3.31495637
-0.06309010	-0.99415166	0.08764765	0.10255071
0.41480465	0.05375654	0.90832116	2.63094218

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [-4.0939114 0.27100848 -1.04295396]

Image 56: 5.jpg

Camera ID: 1

-0.98371946	0.03765702	-0.17572128	-0.75373442
-0.04768340	-0.99744540	0.05318803	0.20507476
-0.17326948	0.06070109	0.98300207	4.72940343

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [0.08777673 -0.05414569 -4.79236805]

Image 57: 50.jpg

Camera ID: 1

-0.90563402	0.09089939	0.41420325	-3.31214883
-0.06354179	-0.99481659	0.07938762	0.10051219
0.41927255	0.04557691	0.90671565	2.63569304

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-4.09828166 0.28093673 -1.02590076]

Image 58: 51.jpg

Camera ID: 1

-0.83566610	0.39486889	0.38176004	-3.58031460
-0.42293751	-0.90608717	0.01139744	0.84519416
0.35040836	-0.15193619	0.92419120	1.48801528

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.15589623 2.40565779 -0.01802264]

Image 59: 52.jpg

Camera ID: 1

-0.83306893	0.38788963	0.39438280	-3.53540849
-0.41877056	-0.90805213	0.00851777	0.94748275
0.36142410	-0.15806002	0.91890677	1.59145253

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.12365038 2.48325702 -0.07616263]

Image 60: 53.jpg

Camera ID: 1

-0.83964863	0.37643429	0.39151935	-3.53590831
-0.41313523	-0.91060936	-0.01048184	0.97290307
0.35257547	-0.17055150	0.92011017	1.57605091

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.12265692 2.48576962 -0.05556613]

Image 61: 54.jpg

Camera ID: 1

-0.83736160	0.38034221	0.39263897	-3.52467565
-0.41630242	-0.90919846	-0.00710349	0.95109619
0.35428499	-0.16940474	0.91966525	1.56993291

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.11168805 2.4712722 -0.05313163]

Image 62: 55.jpg

Camera ID: 1

-0.84025948	0.37800320	0.38868700	-3.56519723
-0.41781052	-0.90831692	-0.01986830	1.01266279
0.34554070	-0.17909204	0.92115562	1.58139760

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-3.11902684 2.55069042 -0.05084758]

Image 63: 56.jpg

Camera ID: 1

-0.84041430	0.37674662	0.38957129	-3.56828725
-0.41530291	-0.90953660	-0.01632991	0.99811347
0.34817711	-0.17551399	0.92085153	1.58966180

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.13780403 2.53116878 -0.05744111]

Image 64: 57.jpg

Camera ID: 1

-0.83327724	0.39241457	0.38943529	-3.50777608
-0.43663768	-0.89919529	-0.02820213	0.85667211
0.33911145	-0.19354231	0.92062196	1.61880698

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.09785064 2.46012562 -0.10009749]

Image 65: 58.jpg

Camera ID: 1

-0.83994712	0.37870443	0.38867954	-3.56757964
-0.42029157	-0.90705879	-0.02448165	1.00708586
0.34328388	-0.18392203	0.92104770	1.53892388

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.10159631 2.5475863 -0.00612196]

Image 66: 59.jpg

Camera ID: 1

-0.84739173	-0.21753529	0.48436108	-3.63396244
0.37993173	-0.88566562	0.26692375	-0.83667634
0.37091662	0.41021312	0.83315428	2.14445217

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.55692279 -2.41121295 0.19681925]

Image 67: 6.jpg

Camera ID: 1

-0.98411072	0.04500237	-0.17175819	-0.77313251
-0.05354474	-0.99753157	0.04542819	0.17174390
-0.16928984	0.05390311	0.98409116	4.74256418

Camera centre C = $-R^T t$ (COLMAP world coordinates): [0.0512159 -0.04952621 -4.80770932]

Image 68: 60.jpg

Camera ID: 1

-0.84596400	-0.21340079	0.48867680	-3.58492444
0.36868798	-0.89616355	0.24690093	-0.91114895
0.38524548	0.38903856	0.83680041	2.19644649

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.54295844 -2.43606656 0.1388456]

Image 69: 61.jpg

Camera ID: 1

-0.84104971	-0.21092385	0.49814307	-3.53167923
0.35719445	-0.90809655	0.21856986	-0.95136523
0.40626041	0.36176205	0.83909517	2.25422246

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.54629674 -2.42433902 0.07571414]

Image 70: 62.jpg

Camera ID: 1

-0.84693156	-0.20595937	0.49019146	-3.55414764
0.35584836	-0.90457961	0.23475024	-0.95452657
0.39506819	0.37325121	0.83940733	2.19352088

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.53704343 -2.4141896 0.12503065]

Image 71: 63.jpg

Camera ID: 1

-0.83714304	-0.20150461	0.50851492	-3.51446553
0.34987354	-0.91187654	0.21463849	-0.96734095
0.42045218	0.35759904	0.83387223	2.33858515

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.58692658 -2.4265523 0.04470557]

Image 72: 64.jpg

Camera ID: 1

-0.83528131	-0.19592496	0.51373004	-3.46325011
0.34470278	-0.91453478	0.21167459	-1.01683900
0.42835165	0.35389200	0.83142968	2.38199433

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.56261204 -2.45144051 0.0139538]

Image 73: 65.jpg

Camera ID: 1

-0.82933619	-0.20569534	0.51951026	-3.42260966
0.35271756	-0.91383236	0.20124797	-0.92007484
0.43334952	0.35014261	0.83042660	2.47292989

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.58561047 -2.41068715 -0.09034273]

Image 74: 66.jpg

Camera ID: 1

-0.82967662	-0.19523519	0.52299132	-3.40030331
0.34244364	-0.91787172	0.20060873	-1.02104075
0.44087306	0.34553542	0.82839376	2.56619391

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-3.60286902 -2.48775418 -0.1426602]

Image 75: 67.jpg

Camera ID: 1

0.26558713	-0.47421297	0.83939594	-3.68210490
0.68142055	0.70825035	0.18451957	-0.53330470
-0.68200404	0.52297562	0.51124064	2.60599847

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.11862594 -2.73126231 1.85685672]

Image 76: 68.jpg

Camera ID: 1

0.27024108	-0.45111857	0.85056557	-3.64641795
0.68965154	0.70715169	0.15593986	-0.48383674
-0.67182625	0.54445250	0.50221605	2.64926128

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.09893395 -2.74521782 1.84646547]

Image 77: 69.jpg

Camera ID: 1

0.27409537	-0.45567951	0.84689310	-3.65984262
0.68651839	0.70940053	0.15950983	-0.47341466
-0.67347177	0.53768678	0.50727576	2.57896859

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.06501632 -2.71855199 1.86676149]

Image 78: 7.jpg

Camera ID: 1

-0.98500178	0.05194441	-0.16453959	-0.80297407
-0.06006343	-0.99719070	0.04475585	0.14057318
-0.16175253	0.05396740	0.98535457	4.78192284

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.00899946 -0.07617966 -4.85030204]

Image 79: 70.jpg

Camera ID: 1

0.26794933	-0.42208476	0.86605289	-3.53989781
0.69260442	0.70925210	0.13137952	-0.40743934
-0.66970312	0.56462900	0.48238140	3.03464658

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.26301984 -2.91860919 1.65541085]

Image 80: 71.jpg

Camera ID: 1

0.26953007	-0.43671664	0.85827275	-3.57008212
0.69237910	0.70732693	0.14247738	-0.45710263
-0.66930167	0.55584818	0.49301935	2.82560172

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.16991274 -2.80639884 1.73615467]

Image 81: 72.jpg

Camera ID: 1

0.26393353	-0.41860772	0.86896874	-3.54940252
0.69311159	0.70883648	0.13094721	-0.43950741
-0.67077225	0.56773095	0.47722757	3.16429777

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.36395715 -2.97073818 1.63178195]

Image 82: 73.jpg

Camera ID: 1

0.26132401	-0.41681533	0.87061745	-3.54132998
0.69386308	0.70813612	0.13075652	-0.42339771
-0.67101698	0.56991948	0.47426574	3.22310086

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.38197001 -3.0131654 1.60989934]

Image 83: 74.jpg

Camera ID: 1

0.26565507	-0.41634375	0.86953163	-3.55846473
0.69245206	0.70994994	0.12837925	-0.40309202
-0.67077383	0.56800437	0.47689989	3.21248648

Camera centre C = $-R^{T}t$ (COLMAP world coordinates): [3.37929797 -3.02007576 1.61391185]

Image 84: 75.jpg

Camera ID: 1

0.63584958	0.07854493	0.76780597	-3.83782601
-0.12099705	0.99265196	-0.00134390	-1.77308369
-0.76226965	-0.09204774	0.64068104	1.64504949

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.47971346 2.21291987 1.89037085]

Image 85: 76.jpg

Camera ID: 1

0.62349263	0.08367909	0.77733824	-3.77403814
-0.10943558	0.99380834	-0.01920490	-1.82055921
-0.77413228	-0.07309435	0.62878965	1.70866205

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.47658148 2.24998856 1.82435151]

Image 86: 77.jpg

Camera ID: 1

0.62040714	0.09081137	0.77900467	-3.77833453
-0.12542034	0.99197863	-0.01575249	-1.72671118
-0.77418649	-0.08793007	0.62682022	1.77282447

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.50003777 2.21186093 1.80489802]

Image 87: 78.jpg

Camera ID: 1

0.62046706	0.09443564	0.77852588	-3.78204106
-0.10920934	0.99345514	-0.03346937	-1.79621245
-0.77659124	-0.06425565	0.62671944	1.66933223

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.4468575 2.24887999 1.83809577]

Image 88: 79.jpg

Camera ID: 1

0.62000997	0.09533525	0.77878035	-3.75376916
-0.11341010	0.99305584	-0.03127690	-1.76238301
-0.77635416	-0.06892957	0.62651650	1.70050905

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.44769955 2.22522661 1.80284279]

Image 89: 8.jpg

Camera ID: 1

-0.98625946	0.05476002	-0.15586407	-0.76731777
-0.06001307	-0.99777054	0.02919549	0.11990654
-0.15391784	0.03814821	0.98734696	4.82344823

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-0.00716373 -0.02234837 -4.88551494]

Image 90: 80.jpg

Camera ID: 1

0.61731122	0.09891976	0.78047533	-3.78803735
-0.10425284	0.99360021	-0.04347382	-1.77313486
-0.77978087	-0.05452989	0.62367322	1.65901210

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.44720951 2.22696467 1.84470331]

Image 91: 81.jpg

Camera ID: 1

0.61580572	0.10068475	0.78143835	-3.75347069
-0.11117285	0.99298232	-0.04033242	-1.70220791
-0.78001533	-0.06203779	0.62267761	1.69162656

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.44166407 2.17312441 1.81111378]

Image 92: 82.jpg

Camera ID: 1

0.61737925	0.09855774	0.78046732	-3.75838292
-0.10358279	0.99366726	-0.04354281	-1.75962016
-0.77981631	-0.05396055	0.62367843	1.65425143

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.4280935 2.208159 1.8249553]

Image 93: 83.jpg

Camera ID: 1

0.60450456	0.01092007	0.79652683	-3.72535203
0.08359429	0.99351566	-0.07706255	-0.26736671
-0.79220340	0.11316976	0.59967189	2.15449832

Camera centre C = $-R^T t$ (COLMAP world coordinates): [3.98114352 0.06249005 1.6547468]

Image 94: 84.jpg

Camera ID: 1

0.59134548	0.01841053	0.80620814	-3.71676475
0.07582444	0.99404085	-0.07831632	-0.17497079
-0.80284567	0.10744228	0.58642560	2.39306984

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.13242488 -0.01476116 1.57942552]

Image 95: 85.jpg

Camera ID: 1

0.60239048	0.01594257	0.79804232	-3.76304934
0.08343917	0.99306522	-0.08282139	-0.26291568
-0.79382846	0.11647880	0.59688279	2.26239006

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.08471216 0.0575646 1.63091589]

Image 96: 86.jpg

Camera ID: 1

0.59106192	0.01717191	0.80644338	-3.75570131
0.07829617	0.99383100	-0.07854720	-0.24909536
-0.80281723	0.10956769	0.58607117	2.32503746

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.10593539 0.05730229 1.6465573]

Image 97: 87.jpg

Camera ID: 1

0.58899820	0.01842147	0.80792436	-3.74868678
0.07549825	0.99411349	-0.07770699	-0.20555589
-0.80459998	0.10676615	0.58414027	2.40546140

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.15892308 0.01658034 1.60755535]

Image 98: 88.jpg

Camera ID: 1

0.59035362	0.01883470	0.80692494	-3.77713790
0.07704605	0.99384771	-0.07956531	-0.21085146
-0.80345909	0.10914205	0.58527045	2.35084936

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.13490359 0.02411896 1.65520765]

Image 99: 89.jpg

Camera ID: 1

0.58970800	0.01435363	0.80748898	-3.77128026
0.07696069	0.99429323	-0.07387847	-0.23732874
-0.80394125	0.10571163	0.58523800	2.32378656

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.11040698 0.04445466 1.66776555]

Image 100: 9.jpg

Camera ID: 1

-0.98657813	0.05799662	-0.15264334	-0.72835804
-0.06343093	-0.99750552	0.03097168	0.05801617

-0.15046632	0.04023829	0.98779591	4.84125814

Camera centre C = $-R^T t$ (COLMAP world coordinates): [0.01354422 -0.09469019 -4.89515087]

Image 101: 90.jpg

Camera ID: 1

0.58809527	0.01299854	0.80868721	-3.75336837
0.07935807	0.99411874	-0.07369005	-0.23769562
-0.80488897	0.10751262	0.58360499	2.33638196

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.1067293 0.03389545 1.65426101]

Image 102: 91.jpg

Camera ID: 1

0.58710585	0.00967823	0.80945232	-3.79199829
0.08292812	0.99394878	-0.07203297	-0.28504546
-0.80525130	0.10941734	0.58275054	2.37351094

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.16121542 0.06031719 1.66574434]

Image 103: 92.jpg

Camera ID: 1

0.58361517	0.00149403	0.81202901	-3.76308116
0.08886222	0.99387508	-0.06569497	-0.33190823
-0.80715354	0.11049928	0.57990781	2.40800877

Camera centre C = $-R^T t$ (COLMAP world coordinates): [4.16931815 0.06941424 1.63750326]

Image 104: 93.jpg

Camera ID: 1

0.99920545	-0.00966488	-0.03866603	2.28920975
0.00963851	0.99995317	-0.00086835	-0.42244293
0.03867261	0.00049498	0.99925181	1.29696274

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-2.33347607 0.44390611 -1.20784455]

Image 105: 94.jpg

Camera ID: 1

0.99953498	0.00022041	-0.03049227	2.26235062
-0.00036193	0.99998919	-0.00463575	-0.40690218
0.03049092	0.00464463	0.99952425	1.35912872

Camera centre C = $-R^T t$ (COLMAP world coordinates): [-2.30288693 0.40008649 -1.29138421]

Image 106: 95.jpg

Camera ID: 1

0.99946846	-0.00345237	-0.03241725	2.24908838
0.00347844	0.99999367	0.00074779	-0.35184024
0.03241446	-0.00086015	0.99947414	1.31348269

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.28924488 0.3607325 -1.23961962]

Image 107: 96.jpg

Camera ID: 1

0.99930136	-0.00335161	-0.03722307	2.31401056
0.00329971	0.99999350	-0.00145557	-0.36072114
0.03722771	0.00133173	0.99930592	1.31914807

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.36031249 0.3667177 -1.23262294]

Image 108: 97.jpg

Camera ID: 1

0.99924977	-0.00083073	-0.03871964	2.14993282
0.00095643	0.99999433	0.00322819	-0.36854830
0.03871673	-0.00326280	0.99924490	1.27629218

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.19738124 0.3744965 -1.19089409]

Image 109: 98.jpg

Camera ID: 1

0.99921614	-0.00477587	-0.03929766	2.14259985
0.00490107	0.99998322	0.00309007	-0.37450187
0.03928224	-0.00328025	0.99922277	1.25477920

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.18837543 0.38884436 -1.16844755]

Image 110: 99.jpg

Camera ID: 1

0.99917080	-0.00431684	-0.04048543	2.12340214
0.00448018	0.99998218	0.00394480	-0.34639571
0.04046768	-0.00412291	0.99917234	1.19769899

Camera centre $C = -R^T t$ (COLMAP world coordinates): [-2.16855761 0.36049392 -1.1093744]

4. Transformation Matrix

The reconstructed point cloud was scaled using a known physical dimension of the photo frame and rotated by 45° about the x-axis. The transformed point cloud was then examined and compared with the physical dimensions of the object.

4.1 Scaling the Point Cloud

The physical dimensions of the photo frame are:

- Width: 17.5 cm
- Height: 23 cm

The scale factor is calculated using a known physical dimension and the corresponding dimension measured in the original reconstructed model:

$$s = \frac{D_{known}}{D_{reconstructed}}$$

where D_{known} is the physical measurement and $D_{\text{reconstructed}}$ is the corresponding measurement in the original model.

- Reference dimension used: 17.5 cm width and 23 cm height
- Known physical dimension: 17.5 cm width and 23 cm height
- Original reconstructed dimension: width = 3.477 PLY units and height = 4.356 PLY units
- Calculated scale factor: 5.1821 cm/PLY unit

The scale factor was applied to the point-cloud coordinates to express the model in centimetres. The scaled point cloud was saved as '*point_cloud_scaled_cm.ply*.'

4.2 Rotation About the X-Axis

A rotation of 45° about the x-axis was applied to the scaled point cloud. The rotation matrix is

$$R_x(45^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 45^\circ & -\sin 45^\circ \\ 0 & \sin 45^\circ & \cos 45^\circ \end{bmatrix}$$

Equivalently,

$$R_x(45^\circ) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.7071 & -0.7071 \\ 0 & 0.7071 & 0.7071 \end{bmatrix}$$

For a point ($p = [x, y, z]^T$), the rotated point is

$$p' = R_x(45^\circ)p$$

The rotation changes the orientation of the point cloud but preserves distances between points.

4.3 Combined Transformation Matrix

For uniform scaling followed by rotation, the combined transformation is

$$T = sR_x(45^\circ)$$

Therefore,

$$T = \begin{bmatrix} s & 0 & 0 \\ 0 & 0.7071s & -0.7071s \\ 0 & 0.7071s & 0.7071s \end{bmatrix}$$

The transformed point is calculated as

$$p' = Tp$$

This equation represents the combined scale and rotation under the stated column-vector convention, without translation.

Scaling matrix (S),

$$S = \begin{bmatrix} 5.183519896688 & 0 & 0 & 0 \\ 0 & 5.183519896688 & 0 & 0 \\ 0 & 0 & 5.183519896688 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

X-axis rotation matrix ($\boldsymbol{R_x(45^\circ)}$),

$$R_x(45^\circ) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.707106781187 & -0.707106781187 & 0 \\ 0 & 0.707106781187 & 0.707106781187 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Combined transformation matrix $\boldsymbol{T = R_xS}$,

$$T = R_xS = \begin{bmatrix} 5.183519896688 & 0 & 0 & 0 \\ 0 & 3.665302069364 & -3.665302069364 & 0 \\ 0 & 3.665302069364 & 3.665302069364 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4.4 Original and Transformed Point-Cloud Comparison

The original point cloud was compared with the scaled and rotated model to examine the effect of the transformations.

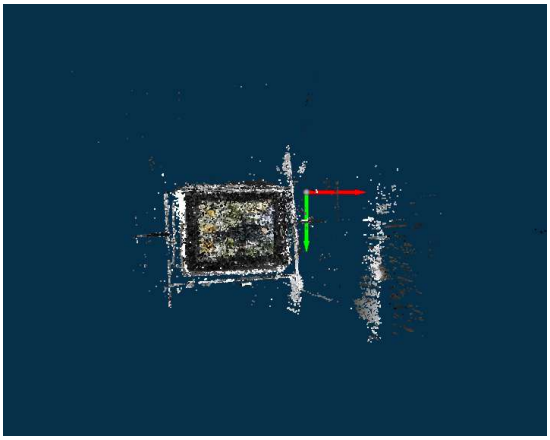


Fig 9. Original reconstructed point cloud

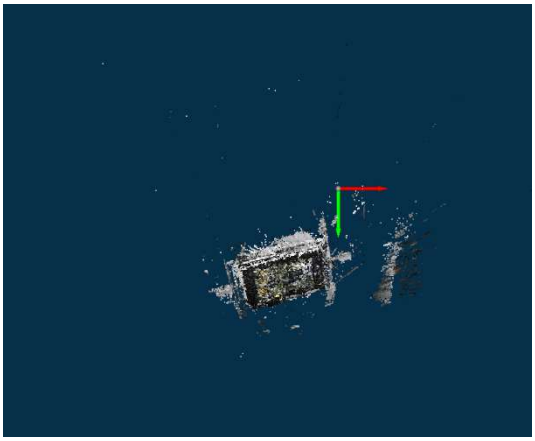


Fig 10. Scaled and rotated point cloud

The comparison illustrates the change in scale and orientation following the transformations. The dimensional measurements provide a separate quantitative assessment of the reconstructed model.

4.5 Dimensional Verification

The dimensions of the transformed model were compared with the physical measurements of the photo frame. The results were exported to '*width_height_comparison.csv'* and visualized in '*width_height_comparison.png'*.

Dimension	Physical measurement	Reconstructed measurement	Absolute error	Percentage error
Width	17.5 cm	18.0240225904509	0.524022590450968	2.99 %
Height	23 cm	22.5817419655848	0.418258034415117	1.82%

Table 3. Dimensional Verification of the Scaled 3D Reconstruction

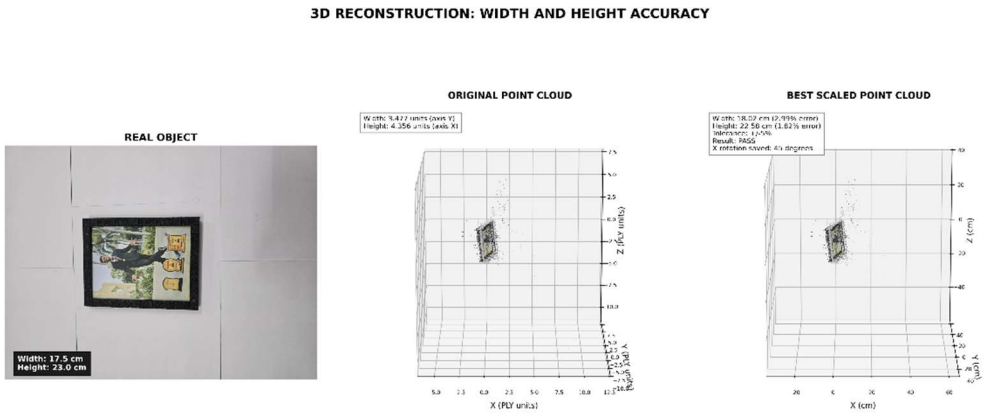


Fig 11. Comparison of physical and reconstructed dimensions

The absolute error is calculated as

$$E_{abs} = \left| D_{reconstructed} - D_{physical} \right|$$

The percentage error is

$$E_{\%} = \frac{\left| D_{reconstructed} - D_{physical} \right|}{D_{physical}} \times 100$$

4.6 Discussion

The scale correction was used to establish the physical scale of the reconstructed photo frame. The 45° rotation about the x-axis changed the model's orientation relative to the coordinate axes.

The dimensional comparison was used to evaluate the agreement between the reconstructed model and the physical measurements. Any differences may be associated with incomplete surface reconstruction, camera pose estimation, point-cloud noise, or the selection of measurement endpoints.

The reconstructed width was 18.024 cm, giving an absolute error of 0.524 cm (approximately 2.99%). The reconstructed height was 22.582 cm, with an absolute error of 0.418 cm (approximately 1.82%). Both dimensions show less than 3% error compared with the physical measurements, indicating good dimensional agreement between the scaled 3D model and the actual photo frame.

5. Conclusion

In this assignment, a three-dimensional reconstruction of a physical photo frame was generated from 110 photographs captured using a Samsung Galaxy S24 Ultra smartphone. COLMAP was used to estimate camera parameters and generate sparse and dense point-cloud models.

The camera calibration results were examined using the intrinsic camera matrix, extrinsic parameters, and estimated camera positions. The reconstructed point cloud was then scaled using a known physical dimension and rotated by 45° about the x-axis.

Finally, the transformed model was compared with the physical measurements of the photo frame to evaluate dimensional accuracy. The final dimensional errors were 0.524 cm (2.99%) for width and 0.418 cm (1.82%) for height. Both dimensions have percentage errors below 3%, indicating good dimensional accuracy of the scaled 3D reconstruction.

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[2] J. L. Schönberger, E. Zheng, J.-M. Frahm, and M. Pollefeys, “Pixelwise View Selection for Unstructured Multi-View Stereo,” *European Conference on Computer Vision (ECCV)*, 2016, pp. 501–518

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GitHub Link:
https://github.com/vasantharan/CV5570_3D_Reconstruction_Assignment**

**Repo will be made public after submission deadline

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